Previous Year Questions

Mains (Functions)

MCQ (Single Correct Answer)

2026

Q1: Given below are two statements:

Statement I : The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=\dfrac{x}{1+|x|}$ is one-one.

Statement II: The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=\dfrac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

(a) Statement I is true but Statement II is false.
(b) Both Statement I and Statement II are false.
(c) Both Statement I and Statement II are true.
(d) Statement I is false but Statement II is true.

Q2: The sum of all the elements in the range of

$f(x)=Sgn(\sin x)+Sgn(\cos x)+Sgn(\tan x)+Sgn(\cot x)$, $x\neq\dfrac{n\pi}{2}, n\in\mathbb{Z}$, where

$Sgn(t)=\begin{cases}1, & if \quad t>0\\ -1, & if \quad t<0\end{cases}$ is:

(a) 4
(b) 0
(c) 2
(d) -2

Q3: If $g(x)=3x^2+2x-3,\ f(0)=-3$ and $4g(f(x))=3x^2-32x+72,$ then $f(g(2))$ is equal to:

(a) $\frac{7}{2}$ (b) $-\frac{25}{6}$

(c) $\frac{25}{6}$ (d) $-\frac{7}{2}$

Q4: Let f be a function such that $3f(x)+2f\left(\frac{m}{19x}\right)=5x, x\neq 0,$ where $m=\sum_{i=1}^{9}(i)^2$. Then $f(5)-f(2)$ is equal to

(a) 36 (b) 9

(c) -9 (d) 18

Q5: Let $f(x)=[x]^2-[x+3]-3, x\in\mathbb{R}$, where $[.]$ is the greatest integer function. Then

(a) $f(x)=0$ for finitely many values of x.

(b) $f(x)<0$ only for $x\in[-1,3)$.

(c) $\int_0^2 f(x)dx=-6$

(d) $f(x)>0$ only for $x\in[4,\infty)$

Q6: Let the domain of the function $f(x)=\log_3\log_5\left(7-\log_2\left(x^2-10x+85\right)\right)+\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha,\beta]$. Then $\alpha+\beta$ is equal to:

(a) 12 (b) 8

(c) 10 (d) 9

Q7: Let f and g be functions satisfying $f(x+y)=f(x)f(y),\ f(1)=7$ and $g(x+y)=g(xy),$ $g(1)=1,$ for all $x,y\in\mathbb{N}$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right)=19607,$ then n is equal to:

(a) 6 (b) 7

(c) 4 (d) 5

Q8: If the domain of the function $f(x)=\sin^{-1}\left(\frac{5-x}{3+2x}\right)+\frac{1}{\log_e(10-x)}$ is $(-\infty,\alpha]\cup[\beta,\gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to

(a) 66 (b) 68

(c) 70 (d) 67

2025

Q1: If the range of the function $f(x)=\frac{5-x}{x^2-3x+2}, x\neq 1,2,$ is $(-\infty,\alpha]\cup[\beta,\infty)$. Then $\alpha^2+\beta^2$ is equal to

(a) 188 (b) 192

(c) 190 (d) 194

Q2: Let the domains of the functions $f(x)=\log_4\log_3\log_7\left(8-\log_2\left(x^2+4x+5\right)\right)$ and $g(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α,β) and $[\gamma,\delta]$ respectively. Then $\alpha^2+\beta^2+\gamma^2+\delta^2$ is equal to:

(a) 15 (b) 13

(c) 16 (d) 14

Q3: Let $f,g:(1,\infty)\to\mathbb{R}$ be defined as $f(x)=\frac{2x+3}{5x+2}$ and $g(x)=\frac{2-3x}{1-x}$. If the range of the function $fog:[2,4]\to\mathbb{R}$ is $[\alpha,\beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(a) 56 (b) 2

(c) 29 (d) 68

Q4: Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x, x\neq 0.$ The $f(3)+f(8)$ is

(a) 13 (b) 11

(c) 10 (d) 12

Q5: If the domain of the function $f(x)=\log_7\left(1-\log_4\left(x^2-9x+18\right)\right)$ is $(\alpha,\beta)\cup(\gamma,o)$, then $\alpha+\beta+\gamma+\hat{o}$ is equal to

(a) 17 (b) 15

(c) 16 (d) 18

Q6: If the domain of the function $f(x)=\log_e\left(\frac{2x-3}{5+4x}\right)+\sin^{-1}\left(\frac{4+3x}{2-x}\right)$ is $[\alpha,\beta)$, then $\alpha^2+4\beta$ is equal to

(a) 4 (b) 3

(c) 7 (d) 5

Q7: If the domain of the function $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ is (a,b), then $(1+a)^2+b^2$ is equal to:

(a) 29 (b) 30

(c) 25 (d) 26

Q8: If the domain of the function $\log_5\left(18x-x^2-77\right)$ is (α,β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ,δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to

(a) 186 (b) 179

(c) 195 (d) 174

Q9: Let $f:[0,3]\to A$ be defined by $f(x)=2x^3-15x^2+36x+7$ and $g:[0,\infty)\to B$ be defined by $g(x)=\frac{x^{2025}}{x^{2025}+1}$, If both the functions are onto and $S=\{x\in\mathbb{Z}; x\in A \text{ or } x\in B\}$ then $n(S)$ is equal to:

(a) 29 (b) 31

(c) 30 (d) 36

Q10: If $f(x)=\frac{2^x}{2^x+\sqrt{2}}$, $x\in\mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to

(a) 82 (b) $81\sqrt{2}$

(c) 41 (d) $\frac{81}{2}$

Q11: Let $f:\mathbb{R}\times\mathbb{R}$ be a function defined by

$f(x)=(2+3a)x^2+\left(\frac{a+2}{a-1}\right)x+b, a\neq 1.$ If $f(x+y)=f(x)+f(y)+1-\frac{2}{7}xy$, then the value of $28\sum_{i=1}^{5}|f(i)|$ is

(a) 735 (b) 675

(c) 715 (d) 545

Q12: The function $f:(-\infty,\infty)\to(-\infty,1)$, defined by $f(x)=\frac{2^x-2^{-x}}{2^x+2^{-x}}$ is :

(a) One-one but not onto.
(b) Onto but not one-one.
(c) Both one-one and onto.
(d) Neither one-one nor onto.

Q13: Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+.....+f\left(\frac{59}{15}\right)\right)$ is equal to

(a) 108 (b) 92

(c) 118 (b) 102

Q14: Let $f(x)=\log_e x$ and $g(x)=\frac{x^4-2x^3+3x^2-2x+2}{2x^2-2x+1}$. Then the domain of fog is

(a) $(0,\infty)$ (b) $[1,\infty)$

(c) $\mathbb{R}$ (d) $[0,\infty)$

Q15: Let $A=\{1,2,3,4\}$ and $B=\{1,4,9,16\}$. Then the number of many-one functions $f:A\to B$ such that $1\in f(A)$ is equal to:

(a) 151 (b) 139

(c) 163 (d) 127

2024

Q1: Let the range of the function $f(x)=\dfrac{1}{2+\sin 3x+\cos 3x}, x\in\mathbb{R}$ be $[a,b]$. If α and β are respectively the A.M. and the G.M. of a and b, then $\dfrac{\alpha}{\beta}$ is equal to

(a) π (b) $\sqrt{\pi}$

(c) $\sqrt{2}$ (d) 2

Q2: If the domain of the function $f(x)=\sin^{-1}\left(\dfrac{x-1}{2x+3}\right)$ is $\mathbb{R}-(\alpha,\beta)$, the $12\alpha\beta$ is equal to

(a) 40 (b) 36

(c) 24 (d) 32

Q3: Let $f(x)=\begin{cases}-a & if \quad -a\le x\le 0\\ x+a & if \quad 0<x\le a\end{cases}$ where $a>0$ and $g(x)=\left(f(|x|)-|f(x)|\right)/2$. Then the function $g:[-a,a]\to[-a,a]$ is

(a) Neither one-one nor onto.
(b) Both one-one and onto.
(c) One-one.
(d) Onto.

Q4: If the function $f(x)=\left(\dfrac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $x=\dfrac{1}{e}$ then :

(a) $e^{\pi}<\pi^{e}$ (b) $e^{2\pi}<(2\pi)^{e}$

(c) $(2e)^{\pi}>\pi^{2e}$ (d) $e^{\pi}>\pi^{e}$

Q5: Let $f(x)=\dfrac{1}{7-\sin 5x}$ be a function defined on $\mathbb{R}$. Then the range of the function $f(x)$ is equal to

(a) $\left[\dfrac{1}{8},\dfrac{1}{5}\right]$ (b) $\left[\dfrac{1}{7},\dfrac{1}{6}\right]$

(c) $\left[\frac{1}{7},\frac{1}{5}\right]$ (d) $\left[\frac{1}{8},\frac{1}{6}\right]$

Q6: The function $f\left(x\right)=\frac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ is

(a) Both one-one and onto.
(b) Onto but not one-one.
(c) Neither one-one nor onto.
(d) One-one but not onto.

Q7: Let $f,g:\mathbb{R}\to\mathbb{R}$ be defined as $f\left(x\right)=\left|x-1\right|$ and $g\left(x\right)=\begin{cases} e^x, & x\geq 0 \\ x+1, & x\leq 0\end{cases}$. Then the function $f\left(g\left(x\right)\right)$ is

(a) neither one-one nor onto.
(b) One-one but not onto.
(c) Both one-one and onto.
(d) Onto but not one-one.

Q8: Let $A=\{1,3,7,9,11\}$ and $B=\left\{2,4,5,7,8,10,12\right\}$. Then the total number of one-one maps $f:A\to B$, such that $f(1)+f\left(3\right)=14$, is :

(a) 120 (b) 180

(c) 240 (d) 480

Q9: If the domain of the function $f(x)=\frac{\sqrt{x^2-25}}{\left(4-x^2\right)}+\log_{10}\left(x^2+2x-15\right)$ is $\left(-\infty,\alpha\right)\cup\left[\beta,\infty\right)$, then $\alpha^2+\beta^3$ is equal to :

(a) 140 (b) 175

(c) 125 (d) 150

Q10: Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be defined as $f\left(x\right)=\begin{cases}\log_e x, & x>0 \\ e^{-x}, & x\leq 0\end{cases}$ and $g(x)=\begin{cases} x, & x\geq 0 \\ e^x, & x<0\end{cases}$.

Then $gof:\mathbb{R}\to\mathbb{R}$ is:

(a) One-one but not onto. (b) neither one-one nor onto.

(c) Onto but not one-one. (d) both one-one and onto.

Q11: If $f(x)=\frac{4x+3}{6x-4}$, $x\neq\frac{2}{3}$ and $(fof)(x)=g(x)$, where $g:\mathbb{R}-\left\{\frac{2}{3}\right\}\to\mathbb{R}-\left\{\frac{2}{3}\right\}$, then $(gogog)(4)$ is equal to

(a) -4

(b) $\frac{19}{20}$

(c) $-\frac{19}{20}$

(d) 4

Q12: If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha,\beta]$, then the value of $5\beta-4\alpha$ is equal to

(a) 9

(b) 12

(c) 11

(d) 10

Q13: If the domain of the function $f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$ is $[-\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to

(a) 11

(b) 12

(c) 9

(d) 8

Q14: If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$, $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(a) $[0,1)$

(b) $[0,3)$

(c) $(0,1]$

(d) $[0,1]$

Q15: Let $f:\mathbb{R}-\left\{\frac{-1}{2}\right\}\to\mathbb{R}$ and $g:\mathbb{R}-\left\{\frac{-5}{2}\right\}\to\mathbb{R}$ be defined as $f(x)=\frac{2x+3}{2x+1}$ and $g(x)=\frac{|x|+1}{2x+5}$. Then the domain of the function fog is

(a) $\mathbb{R}-\left\{-\frac{7}{4}\right\}$

(b) $\mathbb{R}$

(c) $\mathbb{R}-\left\{-\frac{5}{2},-\frac{7}{2}\right\}$ (d) $\mathbb{R}-\left\{-\frac{5}{2}\right\}$

Q16: The functions $f:\mathbb{N}-\{1\}\to\mathbb{N}$; defined by $f(n)=$ the highest prime factor of n, is :

(a) One-one only (b) neither one-one nor onto.

(c) onto only (d) both one-one and onto

2023

Q1: The range of $f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

(a) $[0,2\pi]$ (b) $[0,2\pi)$

(c) $[0,\pi)$ (d) $[0,\pi]$

Q2: For $x\in\mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x)=\sqrt{x}+1$ and $fog(x)=x+3-\sqrt{x}$. Then $f(0)$ is equal to

(a) 5 (b) 0

(c) -3 (d) 1

Q3: Let D be the domain of the function $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g:D\to\mathbb{R}$ defined by $g(x)=x-[x]$, ([x] is the greatest integer function), is (α,β), then $\alpha^2+\frac{5}{\beta}$ is equal to

(a) 45 (b) 136

(c) 46 (d) nearly 135

Q4: The domain of the function $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$ is : (where [x] denotes the greatest integer less than or equal to x)

(a) $(-\infty,-2)\cup[6,\infty)$ (b) $(-\infty,-3]\cup[6,\infty)$

(c) $(-\infty,-2)\cup(5,\infty)$ (d) $(-\infty,-3]\cup(5,\infty)$

Q5: If $f(x)=\dfrac{(\tan 1^o)x+\log_e(123)}{x\log_e(1234)-\tan 1^o}, x>0,$ then the least value of $f(f(x))+f\left(f\left(\dfrac{4}{x}\right)\right)$ is :

(a) 2 (b) 4

(c) 0 (d) 8

Q6: Let the sets A and B denote the domain and range respectively of the function $f(x)=\dfrac{1}{\sqrt{\lceil x\rceil-x}}$,

where $\lceil x\rceil$ denotes the smallest integer greater than or equal to x

Then among the statements

(S1): $A\cap B=(1,\infty)-\mathbb{N}$ and

(S2): $A\cup B=(1,\infty)$

(a) Only (S2) is true (b) Only (S1) is true

(c) Neither (S1) nor (S2) is true (d) Both (S1) and (S2) are true

Q7: Let $f:\mathbb{R}-\{0,1\}\to\mathbb{R}$ be a function such that $f(x)+f\left(\dfrac{1}{1-x}\right)=1+x.$ Then $f(2)$ is equal to

(a) $\dfrac{9}{4}$ (b) $\dfrac{7}{4}$

(c) $\dfrac{7}{3}$ (d) $\dfrac{9}{2}$

Q8: Let $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin 2x\\ \sin^2 x & \cos^2 x & 1+\sin 2x\end{vmatrix}, x\in\left[\dfrac{\pi}{6},\dfrac{\pi}{3}\right].$ If α and β respectively are the maximum and minimum values of f, then

(a) $\alpha^2-\beta^2=4\sqrt{3}$ (b) $\beta^2-2\sqrt{\alpha}=\dfrac{19}{4}$

(c) $\beta^2+2\sqrt{\alpha}=\dfrac{19}{4}$ (d) $\alpha^2+\beta^2=\dfrac{9}{2}$

Q9: Let $f:\mathbb{R}-\{2,6\}\to\mathbb{R}$ be real valued function defined as $f(x)=\dfrac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

(a) $\left(-\infty,-\dfrac{21}{4}\right]\cup[1,\infty)$

(b) $\left(-\infty,-\dfrac{21}{4}\right)\cup(0,\infty)$

(c) $\left(-\infty,-\dfrac{21}{4}\right]\cup[0,\infty)$

(d) $\left(-\infty,-\dfrac{21}{4}\right]\cup\left[\dfrac{21}{4},\infty\right)$

Q10: The absolute minimum value, of the function $f(x)=\left|x^2-x+1\right|+\left[x^2-x+1\right]$, where $[t]$ denotes the greatest integer function, in the interval $[-1,2]$, is :

(a) $\dfrac{3}{4}$

(b) $\dfrac{3}{2}$

(c) $\dfrac{1}{4}$

(d) $\dfrac{5}{4}$

Q11: If the domain of the function $f(x)=\dfrac{[x]}{1+x^2}$, where [x] is the greatest integer $\le x$, is $[2,6)$, then it's range is

(a) $\left(\dfrac{5}{37},\dfrac{2}{5}\right]-\left\{\dfrac{9}{29},\dfrac{27}{109},\dfrac{18}{89},\dfrac{9}{53}\right\}$

(b) $\left(\dfrac{5}{37},\dfrac{2}{5}\right]$

(c) $\left(\dfrac{5}{26},\dfrac{2}{5}\right]$

(d) $\left(\dfrac{5}{26},\dfrac{2}{5}\right]-\left\{\dfrac{9}{29},\dfrac{27}{109},\dfrac{18}{89},\dfrac{9}{53}\right\}$

Q12: The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is :

(a) $\left[2\sqrt{2},\sqrt{1}\right]1$

(b) $\left(\sqrt{5},\sqrt{13}\right)$

(c) $\left(\sqrt{2},\sqrt{7}\right)$

(d) $\left(\sqrt{5},\sqrt{10}\right)$

Q13: Consider the function $f:\mathbb{N}\to\mathbb{R}$, satisfying

$f(1)+2f(2)+3f(3)+....+xf(x)=x(x+1)f(x);\ x\ge 2$ with $f(1)=1$. Then

$\dfrac{1}{f(2022)}+\dfrac{1}{f(2028)}$ is equal to

(a) 8000 (b) 8400

(c) 8100 (d) 8200

Q14: The domain of $f(x)=\dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x}-(2x+3)}, x\in\mathbb{R}$ is

(a) $(-1,\infty)-\{3\}$ (b) $\mathbb{R}-\{-1,3\}$

(c) $(2,\infty)-\{3\}$ (d) $\mathbb{R}-\{3\}$

Q15: Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x)=\dfrac{x^2+2x+1}{x^2+1}$. Then

(a) $f(x)$ is many-one in $(-\infty,-1)$ (b) $f(x)$ is one-one in $(-\infty,\infty)$

(c) $f(x)$ is one-one in $[1,\infty)$ but not in $(-\infty,\infty)$

(d) $f(x)$ is many-one in $(1,\infty)$

Q16: The number of functions $f:\{1,2,3,4\}\to\{a\in\mathbb{Z};|a|\le 8\}$ satisfying $f(n)+\dfrac{1}{n}f(n+1)=1,$ $\forall n\in\{1,2,3\}$ is

(a) 2 (b) 3

(c) 1 (d) 4

Q17: Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\log_{\sqrt{m}}\left\{\sqrt{2}(\sin x-\cos x)+m-2\right\}$, for some m, such that the range of f is $[0,2]$. Then the value of m is

(a) 4 (b) 3

(c) 5 (d) 2

Q18: Let $f(x)=2x^n+\lambda, \lambda\in\mathbb{R}, n\in\mathbb{N}$, and $f(4)=133,\ f(5)=255.$ Then the sum of all the positive integer divisors of $(f(3)-f(2))$ is

(a) 60 (b) 58

(c) 61 (d) 59

Q19: Let $f(x)$ be a function such that $f(x+y)=f(x).f(y)$ for all $x,y\in\mathbb{N}$. If $f(1)=3$ and $\sum_{k=1}^{n} f(k)=3279,$ then the value of n is

(a) 9 (b) 7

(c) 6 (d) 8

Q20: If $f(x)=\frac{2^{2x}}{2^{2x}+3}, x\in\mathbb{R}$, then $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+.....+f\left(\frac{2022}{2023}\right)$ is equal to

(a) 2011 (b) 2010

(c) 1010 (d) 1011

2022

Q1: Let $f(x)=ax^2+bx+c$ be such that $f(1)$, $f(-2)=\lambda$ and $f(3)=4.$ If $f(0)+f(1)+f(-2)+f(3)=14$, then λ is equal to

(a) -4 (b) $\frac{13}{2}$

(c) $\frac{23}{2}$ (d) 4

Q2: Let α,β and γ be three positive real numbers. Let $f(x)=\alpha x^5+\beta x^3+\gamma x, x\in\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be such that $g(f(x))=x$ for all $x\in\mathbb{R}$. If $a_1,a_2,a_3,.....,a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to

(a) 0 (b) 3

(c) 9 (d) 27

Q3: Let $f,g:\mathbb{N}-\{1\}\to\mathbb{N}$ be functions defined by $f(a)=\alpha$, where α is the maximum of the powers of those primes p such that p^{α} divides a, and $g(a)=a+1$, for all $a\in\mathbb{N}-\{1\}$. Then , the function $f+g$

(a) One-one but not onto (b) Onto but not one-one

(c) both one-one and onto (d) neither one-one nor onto

Q4: The number of bijective functions $f:\{1,3,5,7,....,99\}\to\{2,4,6,8,....,100\}$ such that $f(3)\ge f(9)\ge f(15)\ge f(21)\ge .....f(99)$, is

(a) ${}^{50}P_{17}$ (b) ${}^{50}P_{33}$

(c) $33!\times 17!$ (d) $\frac{50!}{2}$

Q5: The total number of functions $f:\{1,2,3,4\}\to\{1,2,3,4,5,6\}$ such that $f(1)+f(2)=f(3)$, is equal to:

(a) 60 (b) 90

(c) 108 (d) 126

Q6: Let the function $f:\mathbb{N}\to\mathbb{N}$ be defined by $f(n)=\begin{cases} 2n, & n=2,4,6,8,..... \\ n-1, & n=3,7,11,15,... \\ \frac{n+1}{2}, & n=1,5,9,13,..... \end{cases}$, then f is

(a) One-one but not onto (b) onto but not one-one

(c) neither one-one nor onto (d) one-one and onto

Q7: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=x-1$ and $g:\mathbb{R}-\{-1,1\}\to\mathbb{R}$ be defined as $g(x)=\frac{x^2}{x^2-1}$. Then the function fog is :

(a) One-one but not onto (b) onto but not one-one.

(c) Both one-one and onto (d) neither one-one nor onto

Q8: Let $f(x)=\frac{x-1}{x+1}$, $x\in\mathbb{R}-\{0,-1,1\}$. If $f^{n+1}(x)=f\left(f^n(x)\right)$ for all $n\in\mathbb{N}$, then $f^6(6)+f^7(7)$ is equal to:

(a) $\frac{7}{6}$ (b) $-\frac{3}{2}$

(c) $\frac{7}{12}$ (b) $-\frac{11}{12}$

Q9: Let $f:\mathbb{N}\to\mathbb{R}$ be a function such that $f(x+y)=2f(x)f(y)$ for natural numbers x and y. If $f(1)=2,$ then the value of α for which $\sum_{k=1}^{10} f(\alpha+k)=\frac{512}{3}\left(2^{20}-1\right)$ holds, is

(a) 2 (b) 3

(c) 4 (d) 6

Q10: Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be two functions defined by $f(x)=\log_e\left(x^2+1\right)-e^{-x}+1$ and $g(x)=\frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ holds?

(a) $(2,3)$ (b) $(-2,-1)$

(c) $(1,2)$ (d) $(-1,1)$

2021

Q1: The range of the function,

$f(x)=\log_{\sqrt{5}}\left(3+\cos\left(\frac{3\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{3\pi}{4}-x\right)\right)$ is

(a) $\left(0,\sqrt{5}\right)$ (b) $[-2,2]$

(c) $\left[\frac{1}{\sqrt{5}},\sqrt{5}\right]$ (d) $[0,2]$

Q2: Let $f:\mathbb{N}\to\mathbb{N}$ be a function such that $f(m+n)=f(m)+f(n)$ for every $m,n\in\mathbb{N}$. If $f(6)=18.$ Then $f(2).f(3)$ is equal to

(a) 6 (b) 54

(c) 18 (d) 36

Q3: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x+y)+f(x-y)=2f(x)f(y)$, $f\left(\frac{1}{2}\right)=-1$, Then, the value of $\sum_{k=1}^{20}\frac{1}{\sin(k)\sin(k+f(k))}$ is equal to:

(a) $\operatorname{cosec}^2(21)\cos(20)\cos(2)$

(b) $\sec^2(1)\sec(21)\cos(20)$

(c) $\operatorname{cosec}^2(1)\operatorname{cosec}(21)\sin(20)$

(d) $\sec^2(21)\sin(20)\sin(2)$

Q4: Consider the function $f:A\to B$ and $g:B\to C$ $(A,B,C\subseteq R)$ such that $(gof)^{-1}$ exists, then

(a) f and g both are one-one

(b) f and g both are onto

(c) f is one-one and g is onto

(d) f is onto and g is one-one

Q5: Let g : N → N be defined as

g(3n + 1) = 3n + 2,

g(3n + 2) = 3n + 3,

g(3n + 3) = 3n + 1, for all n ≥ 0.

Then which of the following statements is true?

(a) There exists an onto function $f:\mathbb{N}\to\mathbb{N}$ such that $fog=f$

(b) There exists a one-one function $f:\mathbb{N}\to\mathbb{N}$ such that $fog=f$

(c) gogog=g

(d) There exists a function $f:\mathbb{N}\to\mathbb{N}$ such that $gof=f$

Q6: Let $f:\mathbb{R}-\left\{\frac{\alpha}{6}\right\}\to\mathbb{R}$ be defined by $f(x)=\frac{5x+3}{6x-\alpha}$. Then the value of α for which $(fof)(x)=x$, for all $x\in\mathbb{R}-\left\{\frac{\alpha}{6}\right\}$, is

(a) No such α exists

(b) 5

(c) 8

(d) 6

Q7: Let $[x]$ denote the greatest integer $\le$x, where $x\in\mathbb{R}$. If the domain of the real valued function $f(x)=\sqrt{\frac{|[x]|-2}{|[x]|-3}}$ is $(-\infty,a]\cup[b,c)\cup[4,\infty)$, $a<b<c$, then the value of $a+b+c$ is

(a) 8

(b) 1

(c) -2

(d) -3

Q8: Let $f:\mathbb{R}-\{3\}\to\mathbb{R}-\{1\}$ be defined by $f(x)=\frac{x-2}{x-3}$. Let $g:\mathbb{R}\to\mathbb{R}$ be given as $g(x)=2x-3$, Then, the sum of all the values of x for which $f^{-1}(x)+g^{-1}(x)=\frac{13}{2}$ is equal to

(a) 3 (b) 5

(c) 2 (d) 7

Q9: The real valued function $f(x)=\frac{\cos ec^{-1}x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to

(a) all real except integers (b) All non-integers except the interval $[-1,1]$

(c) all integers except 0, -1, 1 (d) all real except the interval $[-1,1]$

Q10: If the functions are defined as $f(x)=\sqrt{x}$ and $g(x)=\sqrt{1-x}$, then what is the common domain of the following functions: $f+g,\ f-g,\ f/g,\ g/f,\ g-f$ where $(f\pm g)(x)=f(x)\pm g(x)$, $(f/g)x=\frac{f(x)}{g(x)}$

(a) $0\le x\le 1$ (b) $0\le x<1$

(c) $0<x<1$ (d) $0<x\le 1$

Q11: The inverse of $y=5^{\log x}$ is

(a) $x=5^{\log y}$ (b) $x=y^{\frac{1}{\log 5}}$

(c) $x=5^{\frac{1}{\log y}}$ (d) $x=y^{\log 5}$

Q12: The range of $a\in\mathbb{R}$ for which the function $f(x)=(4a-3)(x+\log_e 5)+2(a-7)\cot\left(\frac{x}{2}\right)\sin^2\left(\frac{x}{2}\right), x\ne 2n\pi, n\in\mathbb{N}$ has critical points, is:

(a) $[1,\infty)$ (b) $(-3,1)$

(c) $\left[-\frac{4}{3},2\right]$ (d) $(-\infty,-1]$

Q13: Let $A=\{1,2,3,....,10\}$ and $f:A\to A$ be defined as $f(k)=\begin{cases} k+1 & if \quad k \quad is \quad odd \\ k & if \quad k \quad is \quad even \end{cases}$. Then the number of possible functions $g:A\to A$ such that $gof = f$ is:

(a) 5^5

(b) 10^5

(c) 5!

(d) ${}^{10}C_5$

Q14: A function $f(x)$ is given by $f(x)=\dfrac{5^x}{5^x+5}$, then the sum of the series

$f\left(\dfrac{1}{20}\right)+f\left(\dfrac{2}{20}\right)+f\left(\dfrac{3}{20}\right)+.....+f\left(\dfrac{39}{20}\right)$ is equal to:

(a) $\dfrac{39}{2}$

(b) $\dfrac{19}{2}$

(c) $\dfrac{49}{2}$

(d) $\dfrac{29}{2}$

Q15: Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions form the set A to the set A$\times$B. Then :

(a) $2y=273x$

(b) $y=91x$

(c) $2y=91x$

(d) $y=273x$

Q16: Let $f,g:\mathbb{N}\to\mathbb{N}$ such that $f(n+1)=f(n)+f(1)\ \forall\ n\in\mathbb{N}$ and g be any arbitrary function. Which of the following statements is NOT true?

(a) If g is onto , then fog is one-one

(b) f is one-one

(c) If f is onto, then $f(n)=n\ \forall\ n\in\mathbb{N}$

(d) If fog is one-one, then g is one-one

Q17: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=2x-1$ and $g:\mathbb{R}-\{1\}\to\mathbb{R}$ be defined as $g(x)=\dfrac{x-\frac{1}{2}}{x-1}$.

Then the composite function $f(g(x))$ is :

(a) One-one but not onto.

(b) onto but not one-one

(c) both one-one and onto

(d) neither one-one not onto

2020

Q1 : Find the number of solutions of $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$, $\|x \in [0, 2\pi]\|$

(a) 2 (b) 4

(c) 6 (d) 8

Q2: Solution set of $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$ contains :

(a) Singleton set (b) two elements

(c) at least four elements (d) infinite elements

Q3: Let $f(x) = \dfrac{x[x]}{x^2+1} : (1,3) \to \mathbb{R}$ then range of $f(x)$ is (where $[f(x)]$ denotes the greatest integer function)

(a) $\left(0, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{7}{5}\right]$ (b) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

(c) $\left(\frac{2}{5}, 1\right) \cup \left(1, \frac{4}{5}\right]$ (d) $\left(0, \frac{1}{3}\right) \cup \left(\frac{2}{5}, \frac{4}{5}\right]$

2017

1. The function f : $R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \dfrac{x}{1+x^2}$, is :

(1) invertible. (2) injective but not surjective.

(3) surjective but not injective. (4) neither injective nor surjective.

2. Let a , b , c ε R. If f(x) = $ax^2 + bx + c$ is such that a + b + c = 3 and f(x+y) = f(x) +f(y) +xy $\forall$ x , y ε R, then $\sum_{n=1}^{10} f(n)$ is equal to :

(1) 330 (2) 165

(3) 190 (4) 255

2016

1. If $f(x)+2f\left(\frac{1}{x}\right)=3x, x\neq 0$ and $S=\{x\in R: f(x)=f(-x)\}$; then S:

(1) contains exactly one element.

(2) contains exactly two elements.

(3) contains more than two elements

(4) is an empty set

2014

1. If $a\in R$ and the equation $-3\left(x-[x]\right)^2+2\left(x-[x]\right)+a^2=0$ (where [x] denotes the greatest integer ≤ x) has no integral solution, then all possible values of a lie in the interval

(1) $(-1,0)\cup(0,1)$

(2) (1,2)

(3) $(-2,-1)$

(3) $\frac{1}{1+\{g(x)\}^5}$

(4) $1+\{g(x)\}^5$

2009

1. For real x, let $f(x)=x^3+5x+1$, then

(1) f is one-one but not onto R

(2) f is onto R but not one-one

(3) f is one-one and onto R

(4) f is neither one-one nor onto R

2008

1. Let f : $N\to Y$ be a function defined as f(x) = 4x +3, where Y = {y ε N : y = 4x +3 for some xεN }. Show that f is invertible and its inverse is

(1) g(y) = $\frac{3y+4}{4}$

(2) g(y) = $4+\frac{y+3}{4}$

(3) g(y) = $\frac{y+3}{4}$

(4) g(y) = $\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+c$

2007

1. The largest interval lying in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ for which the function

$[f(x)=4^{-x^2}+\cos^{-1}\left(\frac{x}{2}-1\right)+\log(\cos x)]$ is defined , is

(1) $[0,\pi]$ (2) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

(3) $\left[-\frac{\pi}{4},\frac{\pi}{2}\right)$ (4) $\left[0,\frac{\pi}{2}\right)$

2006

1. A real valued function f(x) satisfies the functional equation f(x-y) = f(x) f(y) – f(a – x) f(a + y) where a is a given constant and f(0) = 1, f(2a – x) is equal to

(1) -f(x) (2) f(x)

(3) f(a) + f(a-x) (4) f(-x)

2005

1. Let f : $(-1,1)\to B$, be a function defined by $f(x)=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then f is both one-one and onto when B is the interval

(1) $\left(0,\frac{\pi}{2}\right)$ (2) $\left[0,\frac{\pi}{2}\right)$

(3) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (4) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

2004

1. The range of the function f(x) = ${}^{7-x}P_{x-3}$ is

(1) { 1, 2, 3} (2) {1,2,3,4,5}

(3) {1,2,3,4} (4) {1,2,3,4,5,6}

2. If f : $R\to S$, defined by $f(x)=\sin x-\sqrt{3}\cos x+1$, is onto, then the interval of S is

(1) [0,3] (2) [-1,1]

(3) [0,1] (4) [-1,-3]

3. The graph of the function $y = f(x)$ is symmetrical about the line x = 2 , then

(1) f(x + 2) = f(x -2) (2) f(2+x) = f(2-x)

(3) f(x) = f(-x) (4) f(x) = -f(-x)

4. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is

(1) [2,3] (2) [2,3)

(3) [1,2] (4) [1,2)

2003

1. A function f from the set of natural numbers to integers defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & when\, n\, is\, odd \\ -\frac{n}{2}, & when\, n\, is\, even \end{cases}$$ is

(1) one-one but not onto (2) onto but not one-one

(3) one-one and onto both (4) neither one-one nor onto

2. If $f : R \to R$ satisfies f(x+y) = f(x) + f(y), for all x,y ε R and f(1) = 7, then $\sum_{r=1}^{n} f(r)$ is

(1) $\frac{7n}{2}$ (2) $\frac{7(n+1)}{2}$

(3) 7n(n+1) (4) $\frac{7n(n+1)}{2}$

3. Domain of definition of the function f(x) = $\frac{3}{4-x^2} + \log_{10}\left(x^3 - x\right)$, is

(1) (1,2) (2) $(-1,0) \cup (1,2)$

(3) $(1,2)\cup(2,\infty)$ (4) $(-1,0)\cup(1,2)\cup(2,\infty)$

4. The function f(x) = $\log\left(x+\sqrt{x^2+1}\right)$, is

(1) an even function (2) an odd function

(3) a periodic function (4) neither an even nor an odd function

2002

1. The period of $\sin^2\theta$ is

(1) π^2 (2) π

(3) π^3 (4) $\pi/2$

2. Which one is not periodic

(1) $|\sin 3x|+\sin^2 x$ (2) $\cos\sqrt{x}+\cos^2 x$

(3) cos4x + $\tan^2 x$ (4) $\cos 2x+\sin x$

3. If f(x+y) = f(x).f(y) $\forall$ x,y and f(5) = 2, f'(0) = 3, then f'(5) is

(1) 0 (2) 1

(3) 6 (4) 2

4. The domain of $\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$ is

(1) [1,9] (2) [-1,9]

(3) [-9,1] (4) [-9,-1]

Assertion – Reason Type

1. Let f be a function defined by $f\left(x\right)=\left(x-1\right)^{2}+1,\left(x\geq 1\right)$

   Statement – I : The set { x : f(x) = $f^{-1}(x)$ } = {1,2}.

   Statement – II : f is bijection and $f^{-1}\left(x\right)$ = $1+\sqrt{x-1}$, $x\geq 1$.

2. Let f(x) = $\left(x+1\right)^{2}-1$, $x\geq -1$

   Statement – I : The set { x : f(x) = $f^{-1}(x)$ } = {0,-1}

   Statement – II : f is a bijection.